

RTC DS3231 module. Students can use push-button switches (S1, S2, S3, or S4) to select meal schedules based on their daily routine, including full-day meal plans for holidays or specific meal reminders at home. The system supports up to four meal times, displaying their names (e.g., breakfast, lunch, or dinner).

The system triggers an alert at each scheduled time by turning on the buzzer and LED while displaying the meal name on the LCD screen. The student must press the push-to-on switch S5 to acknowledge the alert, which turns off the buzzer and LED. At this point, the SIM900A GSM module sends a predefined message to a designated mobile number (e.g., the parent's).

If no response is received, the student may have missed the alert. In such cases, parents can call to check on their child.

Software

Before proceeding, the 'LiquidCrystal.h', 'LiquidCrystal_I2C', and 'RTC' libraries must be installed. The display should then be configured in the code according to the setup, and the necessary pins should be defined. The function should be modified to customise the message for scheduling. Alternatively, selecting the Arduino Uno board, choosing the correct port, and uploading the code is sufficient.

The message and phone number must be configured in the sendSMS function in the source code. The code can be uploaded after selecting the Arduino Uno board and the correct port number based on the laptop connection. Fig. 3 shows a snippet of the source code.

Construction and testing

After uploading the source code to the Arduino Uno, the components should be assembled as shown in Fig. 2. The 12V, 2A adaptor should be connected to the Arduino Uno's DC socket.

```
AL_EFY-DAILY_MEAL_REMAINDER_SYSTEM-RAKESH_JAIN.ino
1 //RAKESH JAIN PROGRAM FOR AUTOMATIC DAILY MEAL REMAINDER GADGET
2 #include <Wire.h>
3 #include <RTCLib.h>
4 ///////////////Spi Display//////////
5 // #include <LiquidCrystal.h>. ///uncomment for spi
6 //LiquidCrystal lcd(12, 11, 5, 4, 3, 2); ///uncomment for spi
7 ///////////////Spi Display//////////
8
9 #include <LiquidCrystal_I2C.h> ///I2C comment for SPI
10 LiquidCrystal_I2C lcd(0x27, 16, 2); // I2C Address 0x27 (Change if needed)
11
12
13 #define BUZZER_PIN 13
14 #define BUTTON_1 9
15 #define BUTTON_2 8
16 #define BUTTON_3 7
17 #define BUTTON_4 6
18 #define BUTTON_5 A0
19 RTC_DS3231 rtc;
```

Fig. 3: Snippet of the source code

Testing process. Once powered on, the LCD should display the current date, time, and year retrieved from the RTC module. The buzzer remains off initially.

On pressing S1 momentarily →LCD briefly displays 'SCHEDULE-1' before returning to the time display.

Alert: 7:30 AM – 'BREAKFAST'

On pressing S2 momentarily →LCD briefly displays 'SCHEDULE-2' before returning to the time display.

Alerts:

7:30 AM – 'BREAKFAST'

12:00 PM – 'LUNCH'

On pressing S3 momentarily →LCD briefly displays 'SCHEDULE-3' before returning to the time display.

Alerts:

7:30 AM – 'BREAKFAST'

12:00 PM – 'LUNCH'

4:30 PM – 'EVENING SNACKS'

On pressing S4 momentarily

→LCD briefly displays 'SCHEDULE-4' before returning to the time display.

Alerts:

7:30 AM – 'BREAKFAST'

12:00 PM – 'LUNCH'

4:30 PM – 'EVENING SNACKS'

8:00 PM – 'DINNER'

Stopping the alarm. Press S5 while the buzzer is on and the →The buzzer turns off immediately, and the LCD displays the date, time, and year.

The SIM900A module sends an SMS to the pre-set phone number with the message: 'SCHEDULE FOLLOW'.

After stopping the alarm, it will activate again at the next scheduled meal time.

Missed alarm alert. If S5 is not pressed, it is assumed the student has not followed the meal schedule. The parents understand that the schedule is not followed if they do not receive any message. **EFY**

TO DOWNLOAD SOURCE CODE AND RESOURCES



Scan this QR code or go to <https://sourcecode.electronicsforu.com> to access and download source code and other resources for this project

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